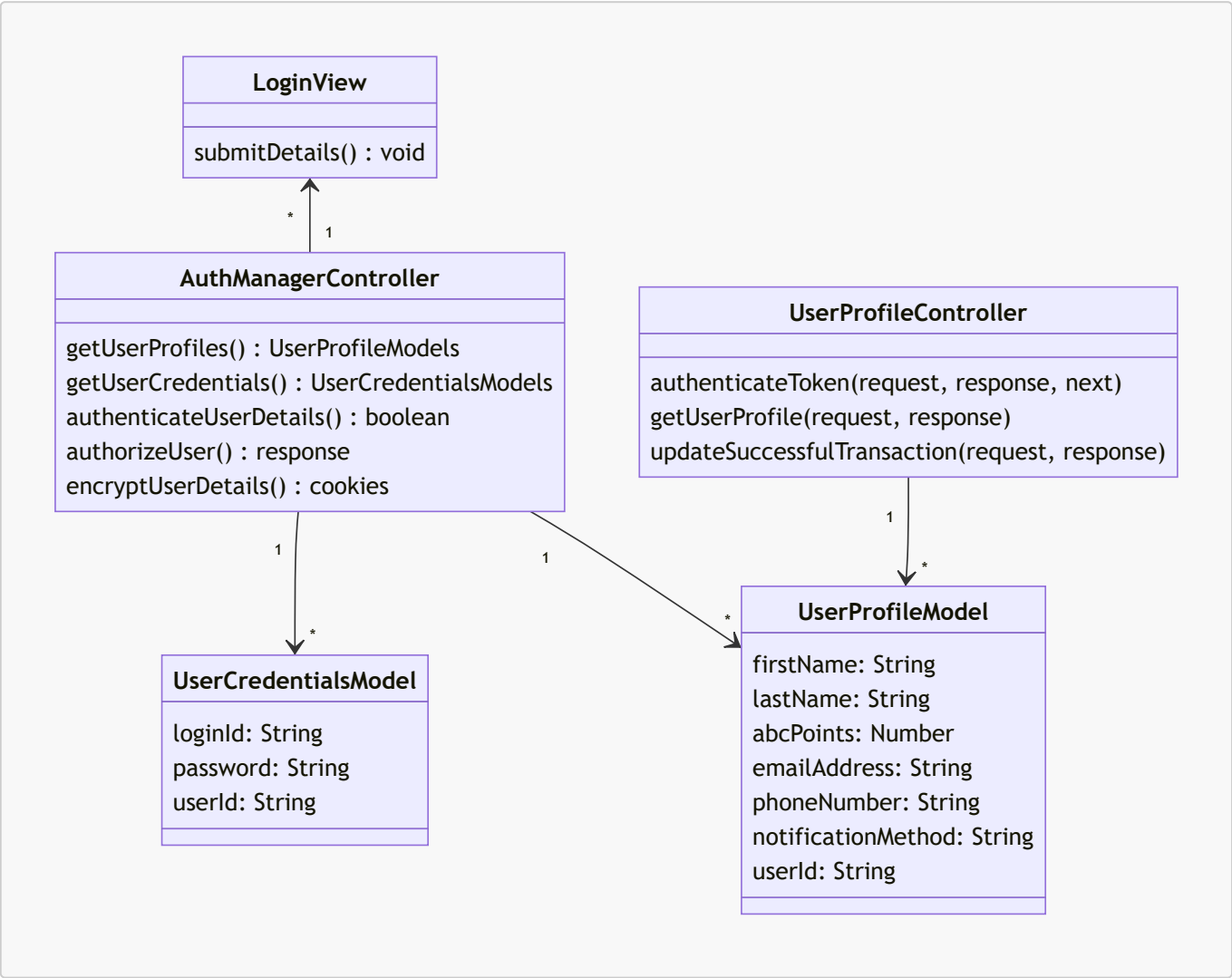


Bank App Class Diagrams

Login (Server)

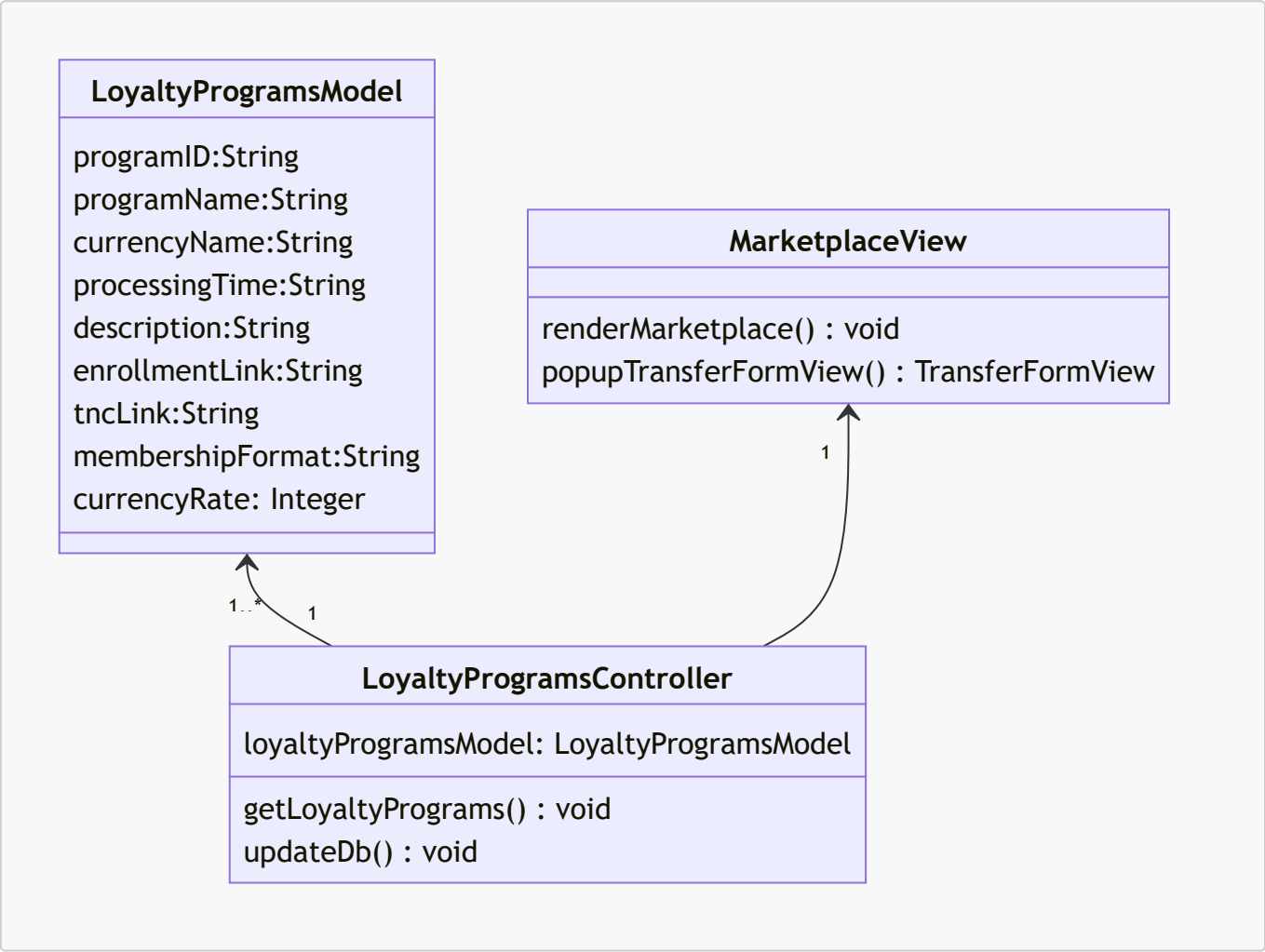
This component describes user authentication for logging in to the bank application, access API routes. This component also describes the encryption and decryption of crucial user credentials and their respective user profile details and services the frontend to provide protection and user authorization to various routes.



Bank App Class Diagrams

Query Loyalty Program Details (Loyalty Points Marketplace)

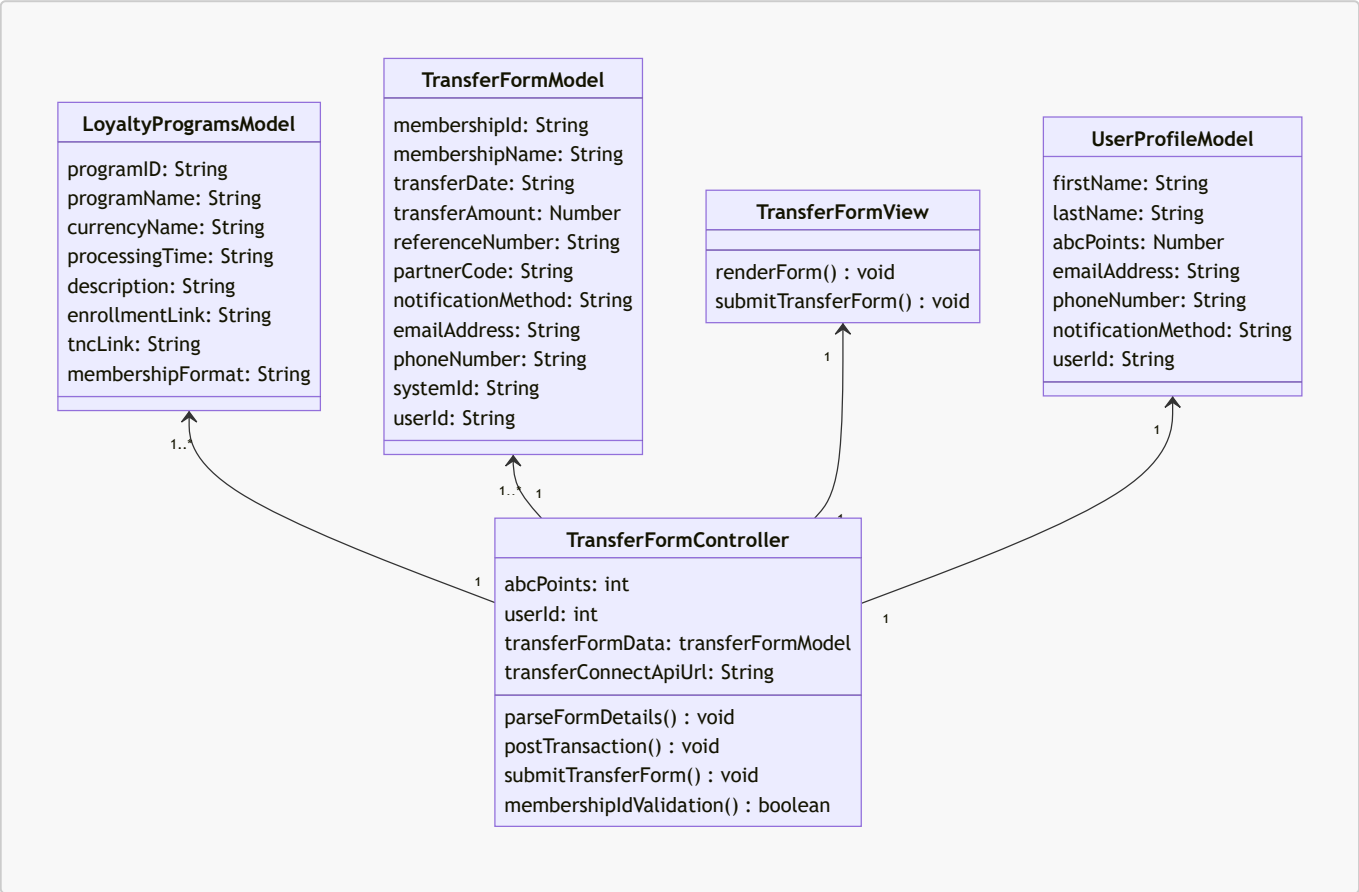
This component queries Loyalty Program details from the TransferConnect API endpoint daily to get updated Loyalty Program details and updates the Loyalty Program details in the bank application database. This component also services the marketplace view to bank users, to show the available Loyalty Program partners, currency rates.



Bank App Class Diagrams

Submit Credit Transfer Request

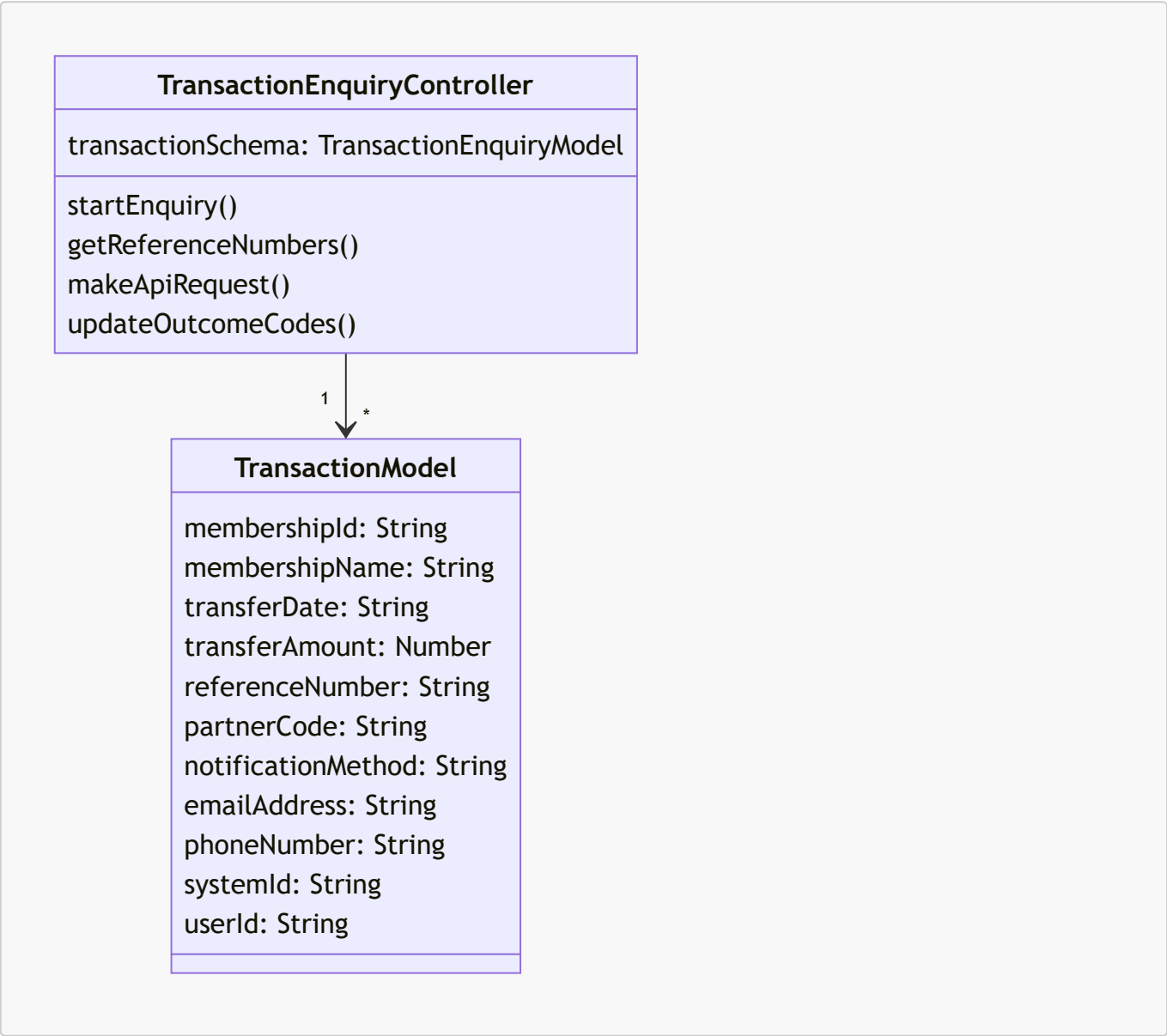
This component facilitates credit transfers by validating user form inputs, and generating unique reference numbers for each transaction, lints transaction details and sends a post request to the TransferConnect API endpoint to submit transactions. This component also serves the bank application frontend, providing forms for various Loyalty Programs and their currency rates to allow users to submit inputs on the website.



Bank App Class Diagrams

Enquire Transaction Status

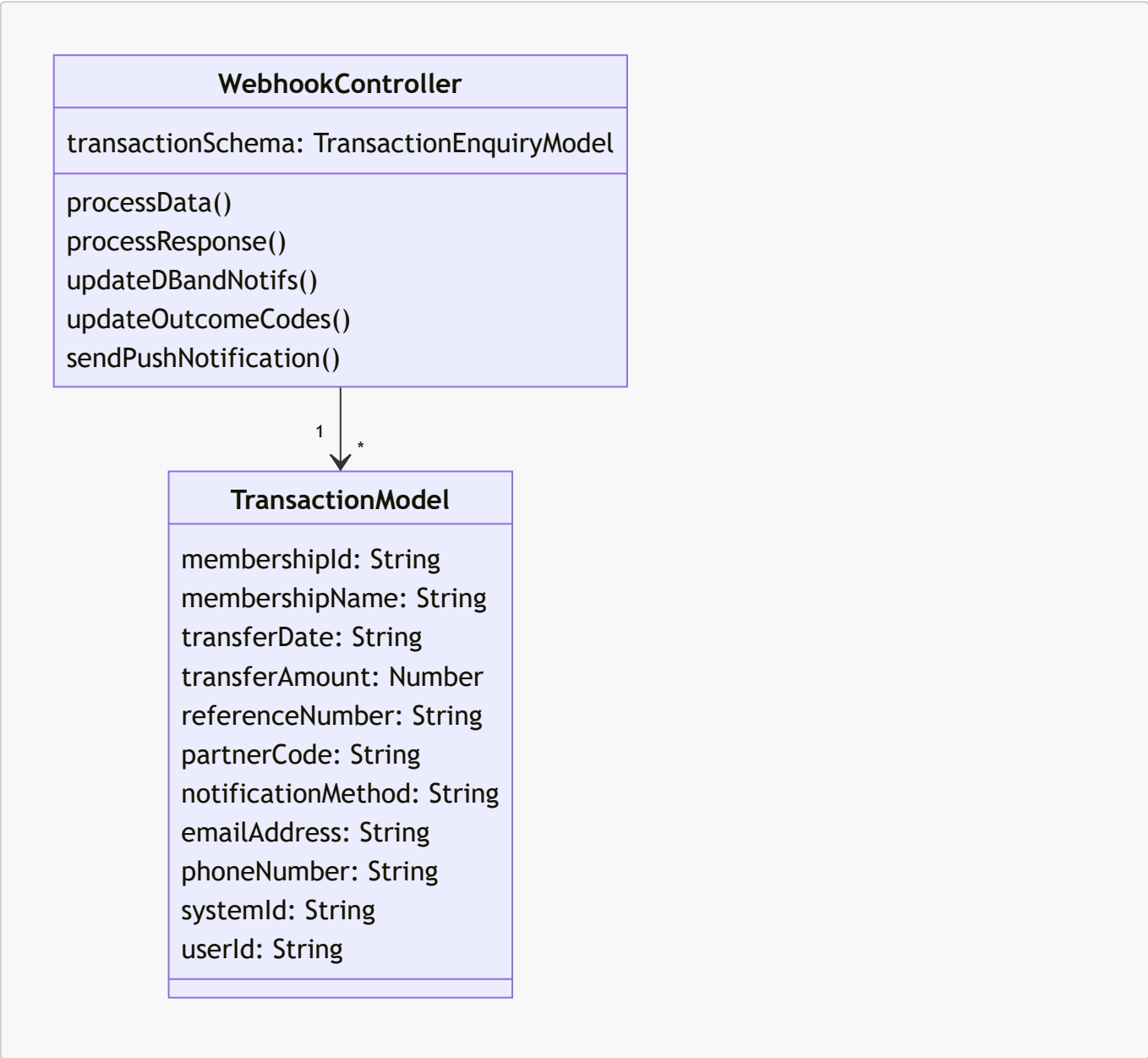
This component in the bank application sends queries to the TransferConnect API endpoint to enquire about transaction status updates. It also processes the updated transactions and updates the bank application database if there are any updates.



Bank App Class Diagrams

Receive Transaction Status Update Webhook Post

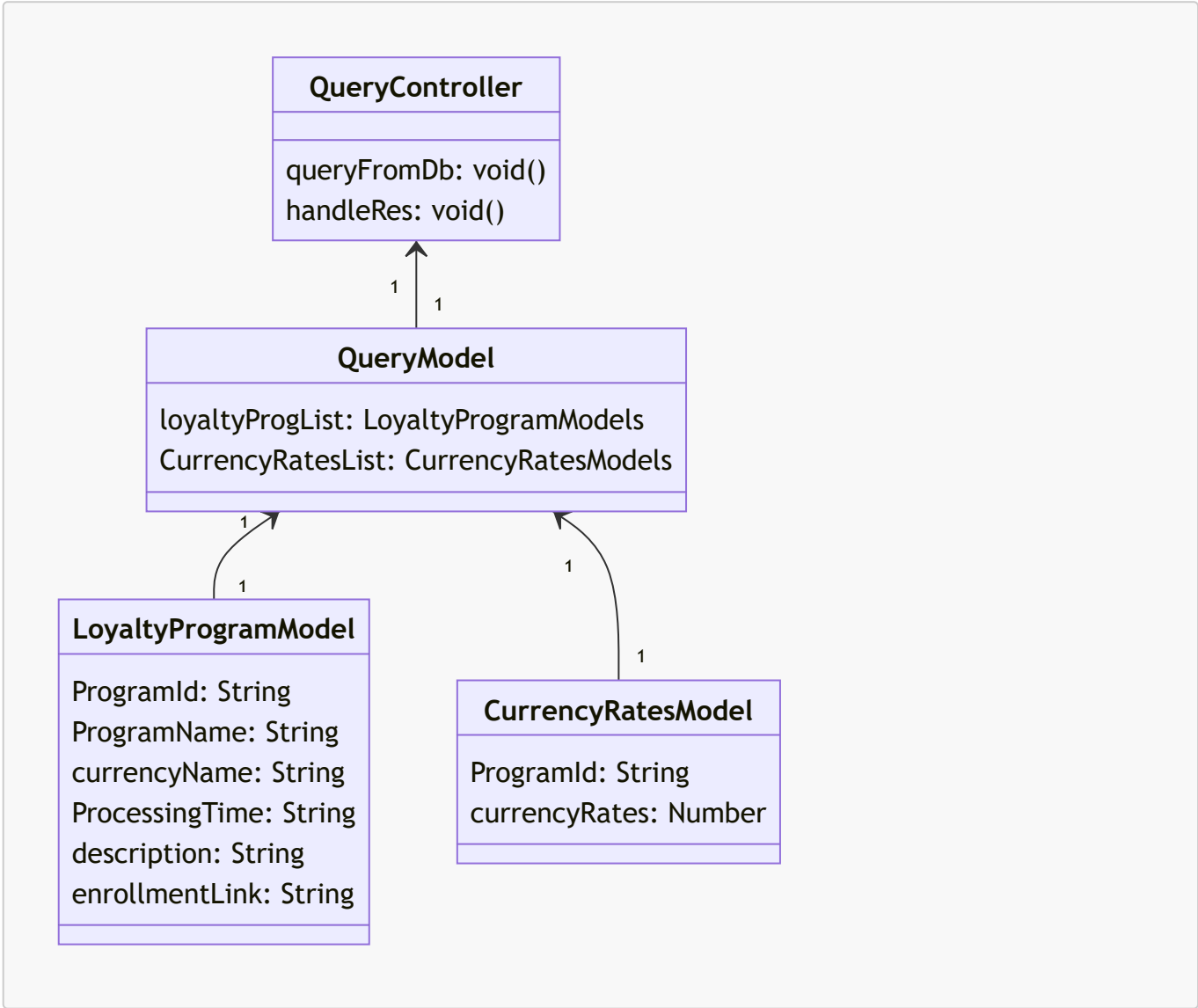
This component describes the functionalities to support the Bank App in handling transaction status updates from TransferConnect via post. Upon receiving this request, this component also processes the response, updates the transaction details to the database.



TransferConnect Class Diagrams

Loyalty Program Query

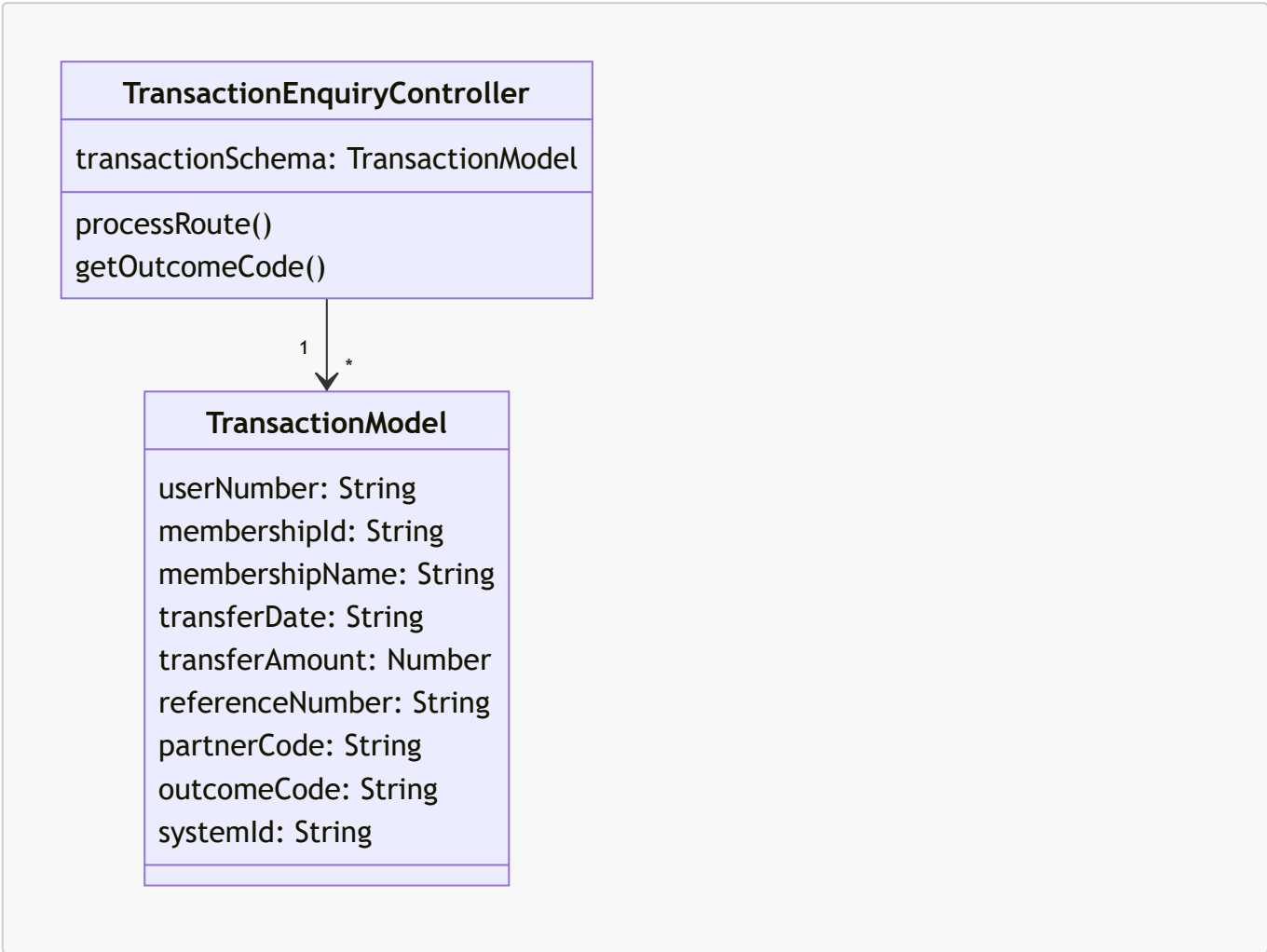
This component handles Loyalty Program details queries from various bank applications to give the various bank applications updated Loyalty Program information daily. It also serves to compile mapped exchange rates from each Loyalty Programs to each bank and be the ground truth for exchange rates.



TransferConnect Class Diagrams

Enquire Transaction Status

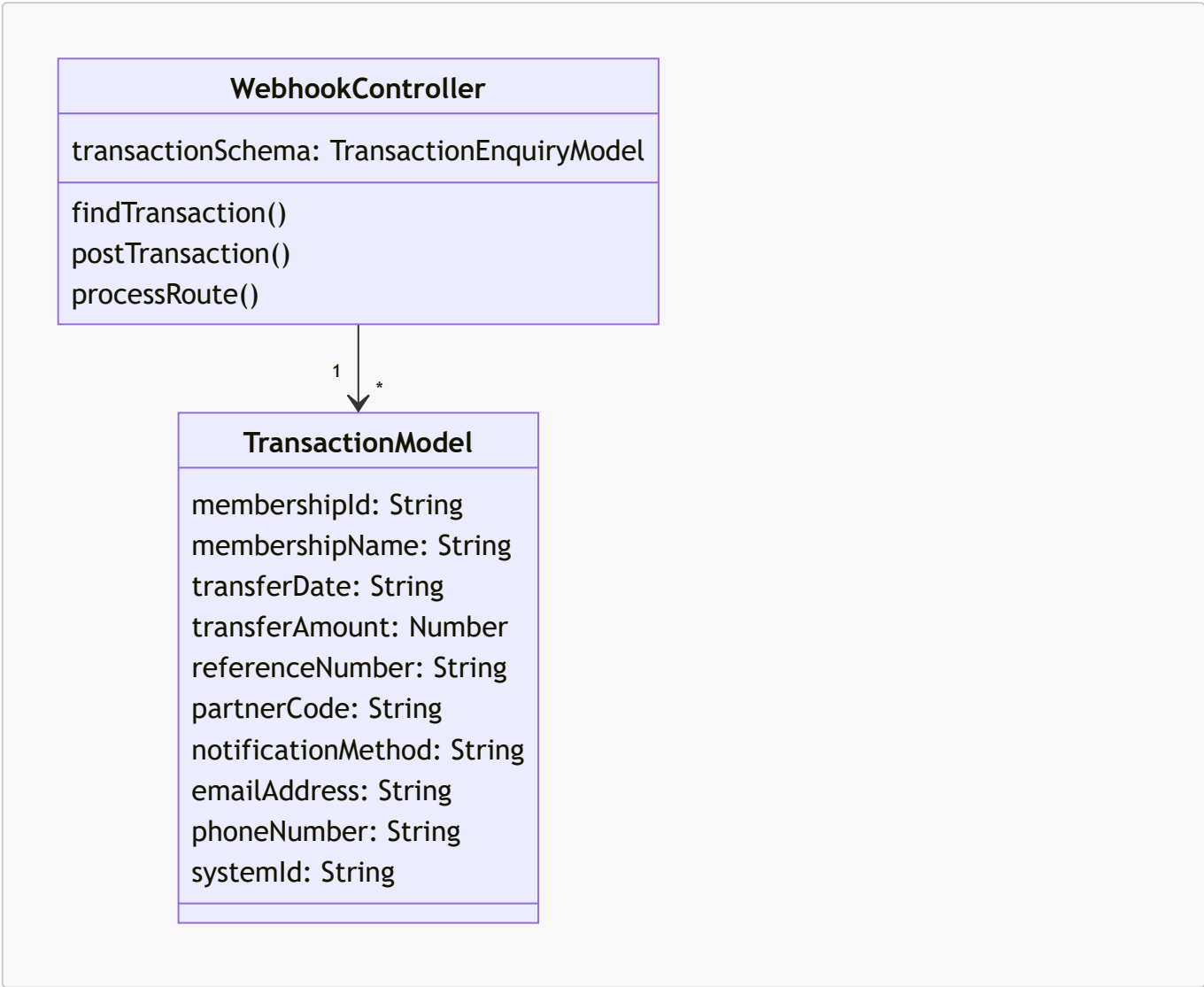
This component handles transaction status queries from various bank applications to give banks updates about their transaction status.



TransferConnect Class Diagrams

Post Transaction Status Update Webhook

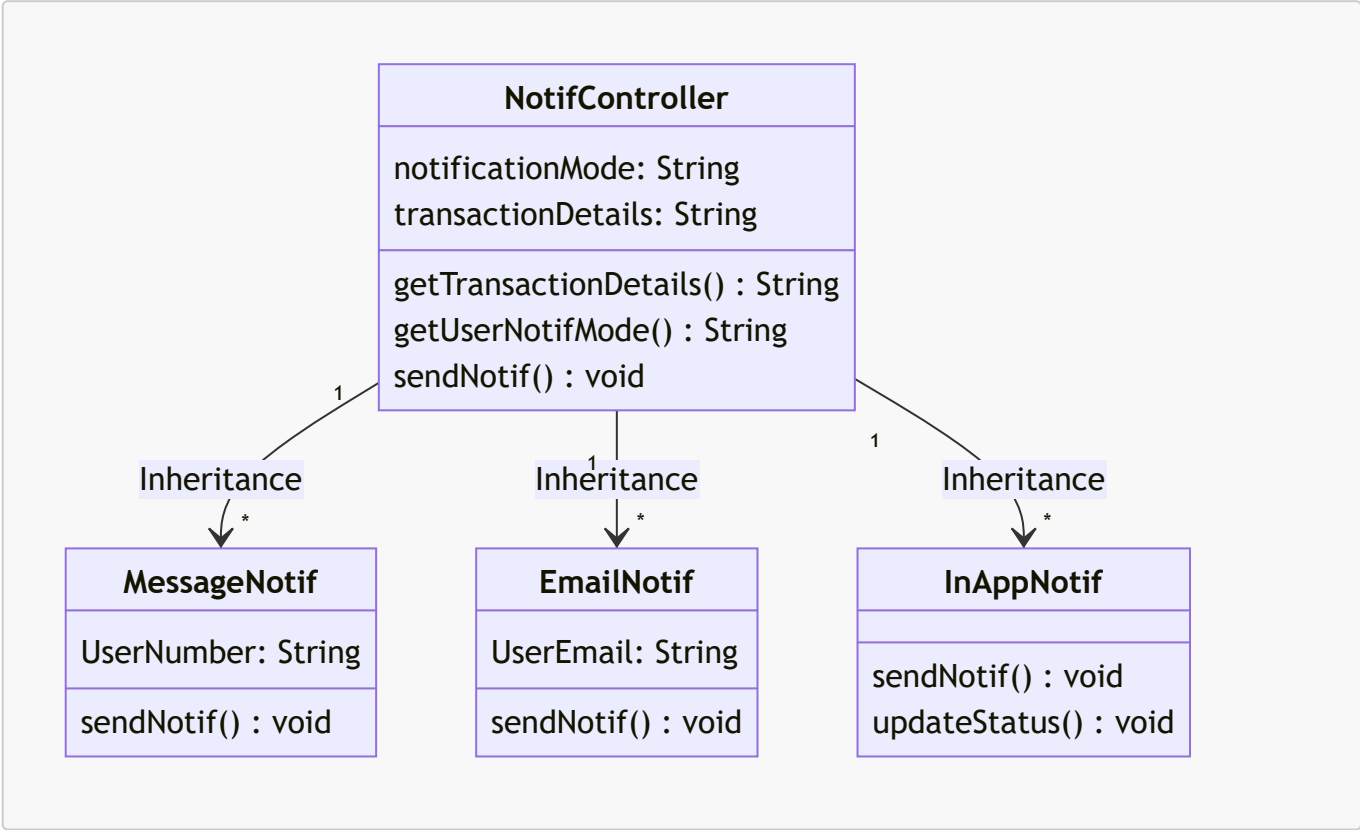
This component describes transaction status update processes. It serves to post updated transaction details to various bank applications via webhook.



TransferConnect Class Diagrams

Notify Transaction Status

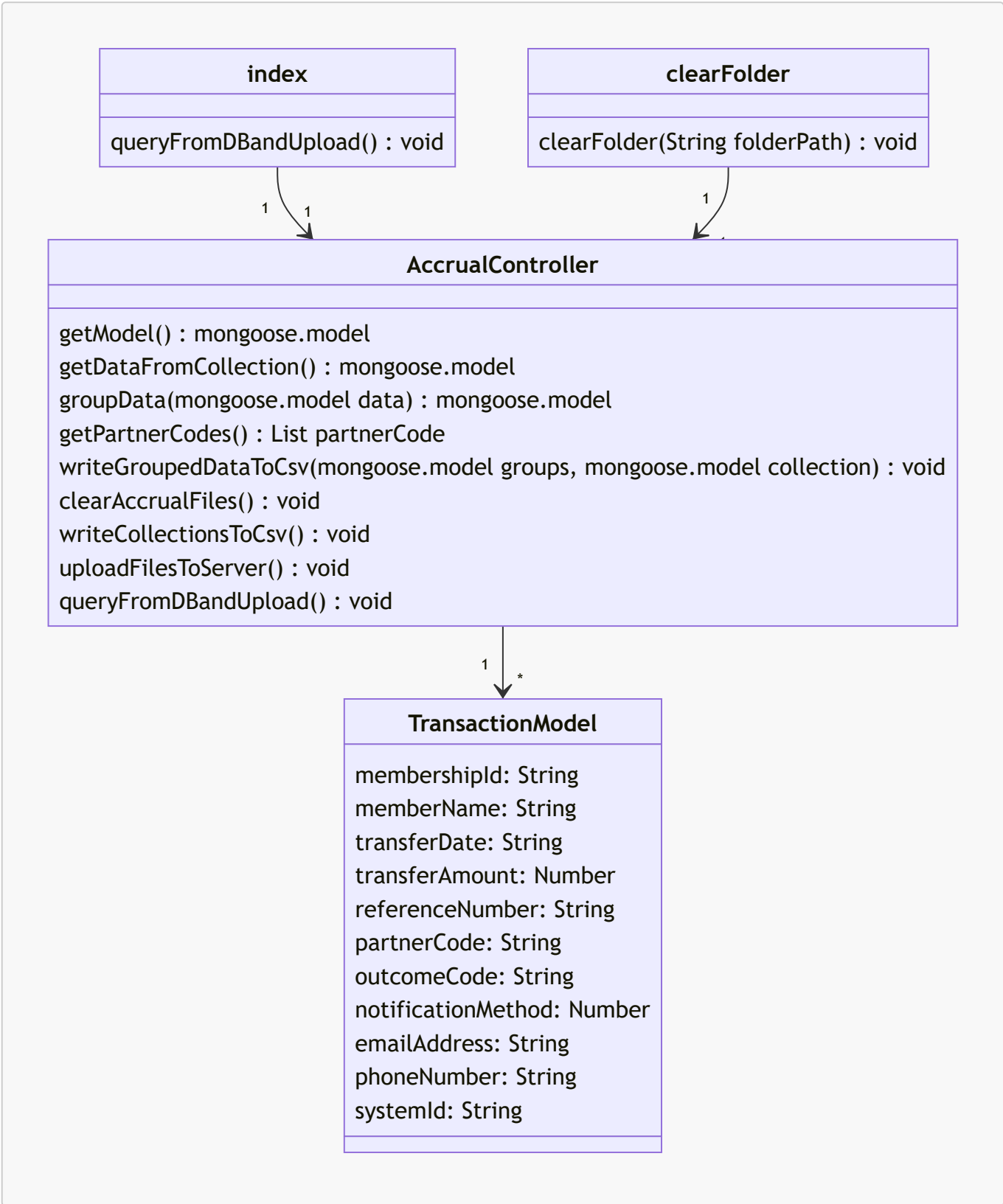
Upon transaction status updates after processing of the transaction handback files, retrieves user profile details, their notification details, and directly notifies transaction users their updated transaction status.



TransferConnect Class Diagrams

Send Transfer Fulfilment Accrual Files

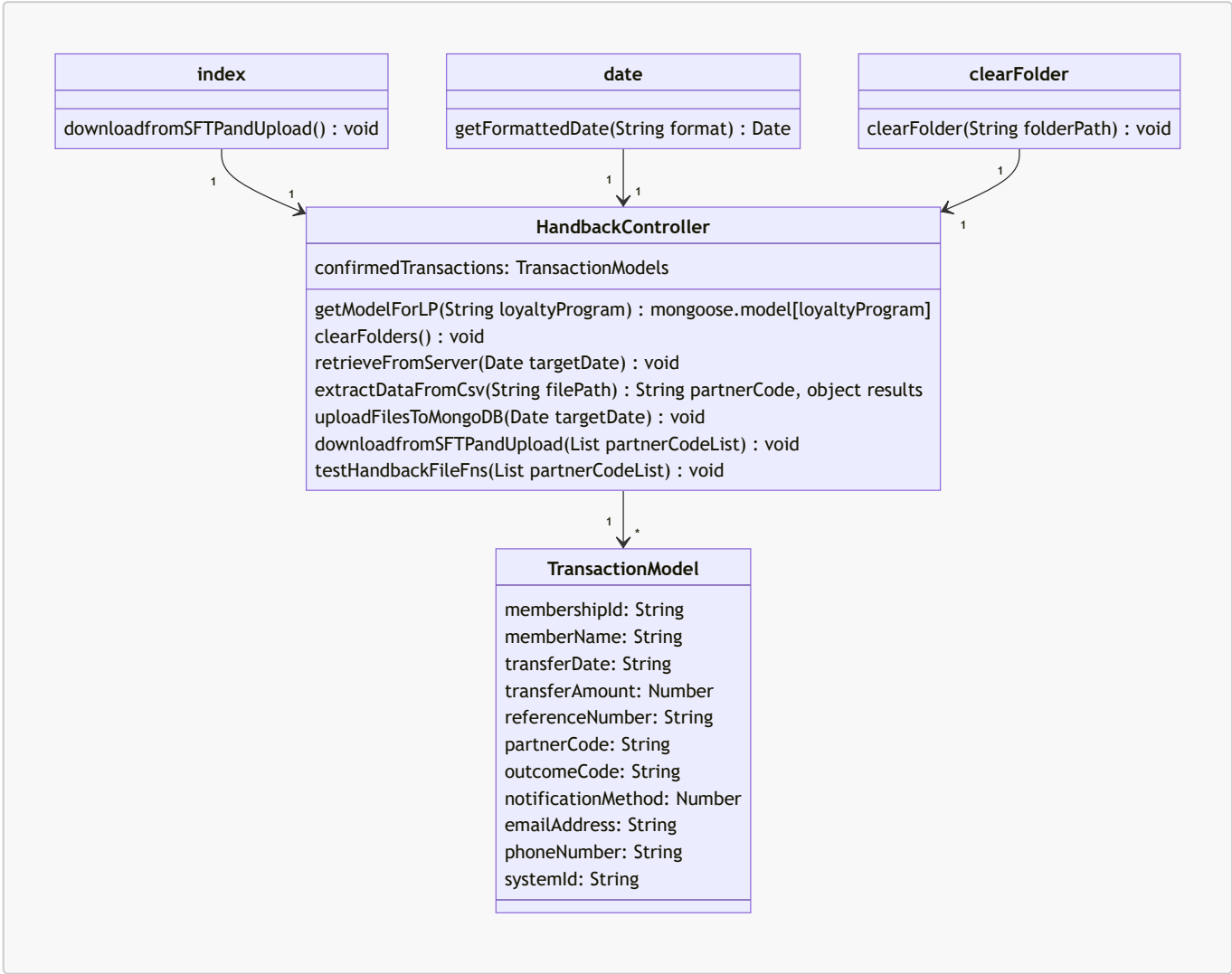
This component describes several functions to support the querying of, compilation of and sending of transaction details to the Loyalty Program providers for the day.



TransferConnect Class Diagrams

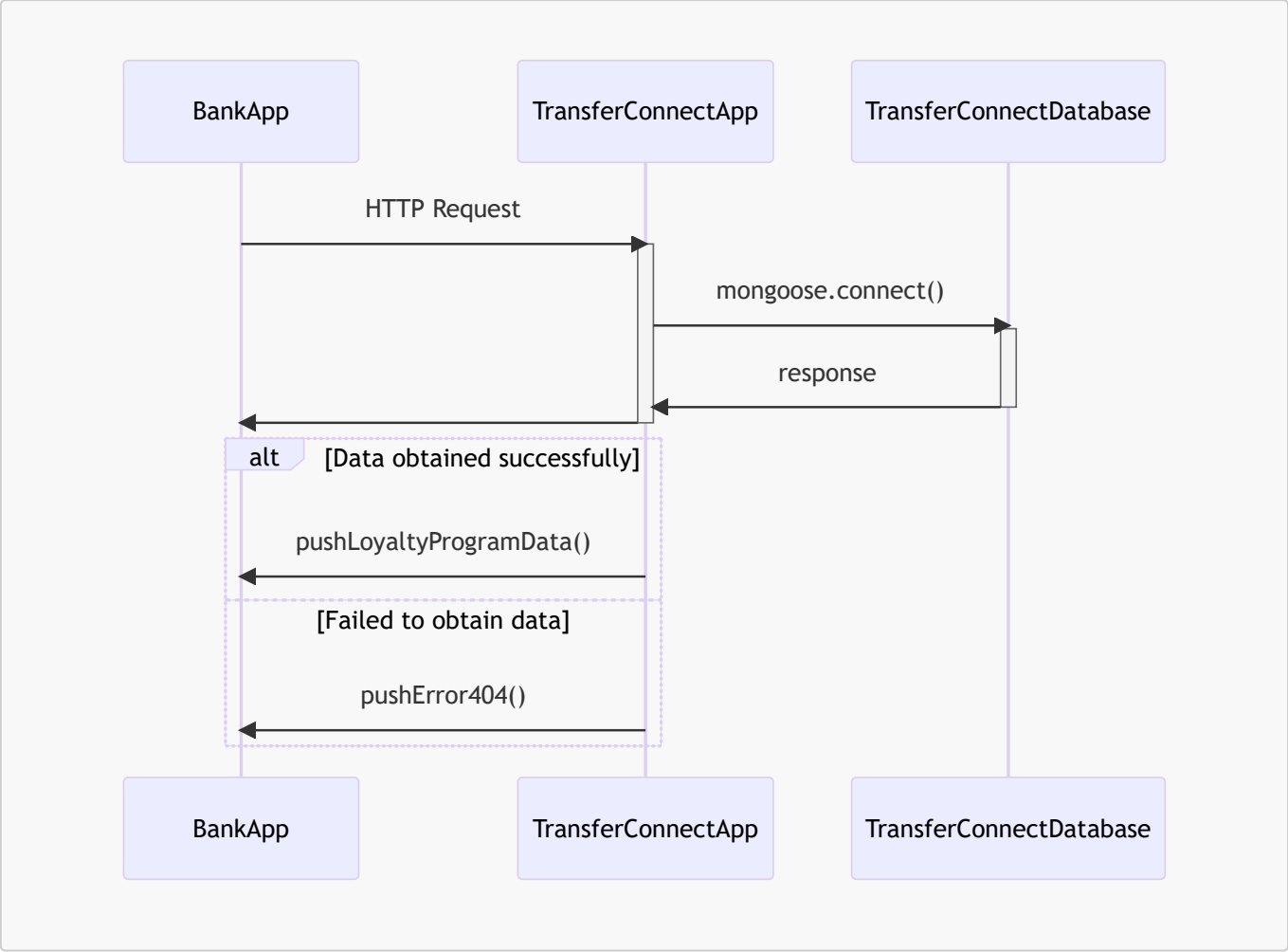
Receive Transfer Fulfilment Handback File

This component describes several functions to support the processing of transaction handback files. The querying of, downloading transaction handback files from Loyalty Program providers when they have processed them and updating transaction details in the TransferConnect database.



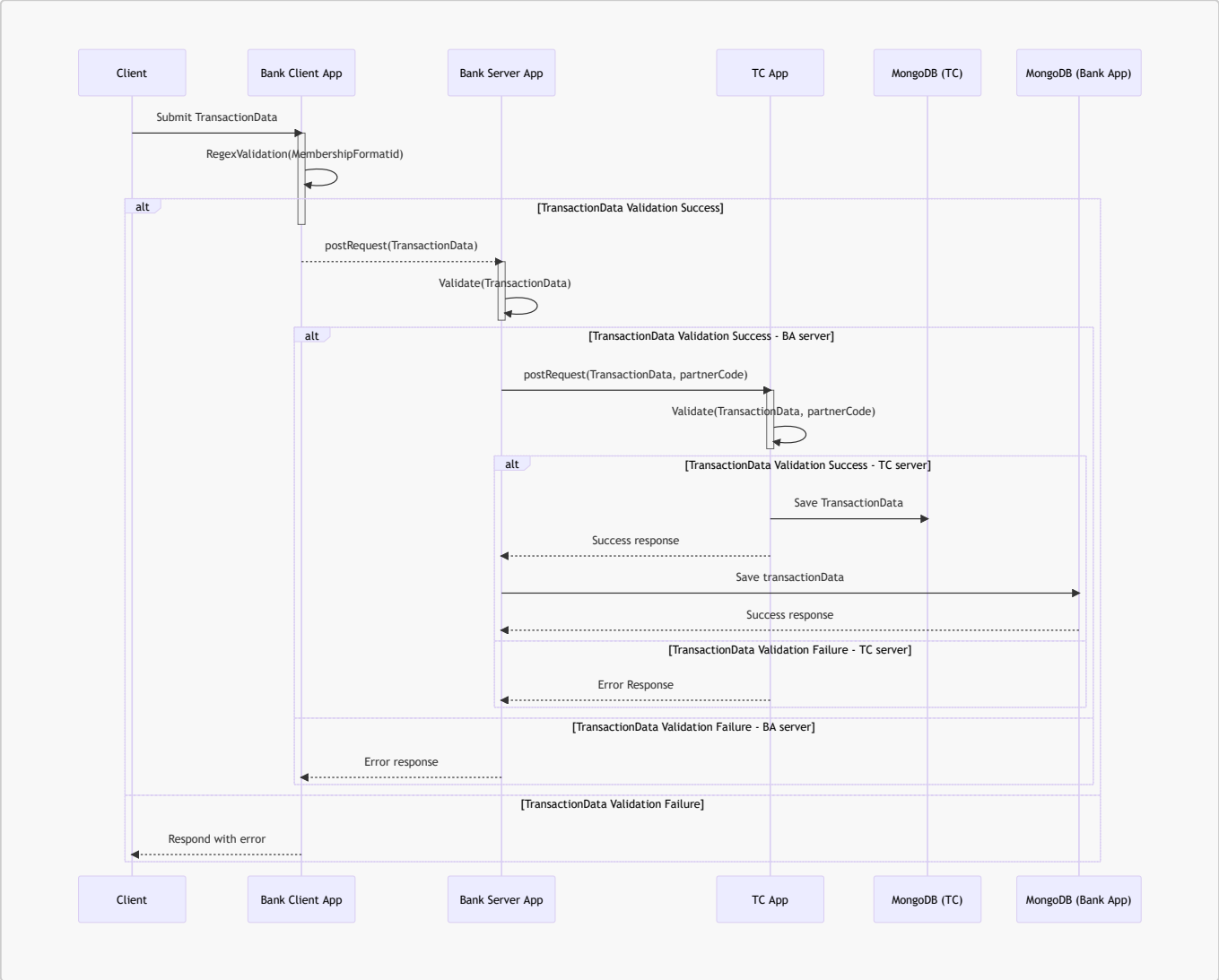
Sequence Diagrams

Daily interactions between TransferConnect App and Bank App to supply information about Loyalty Programs



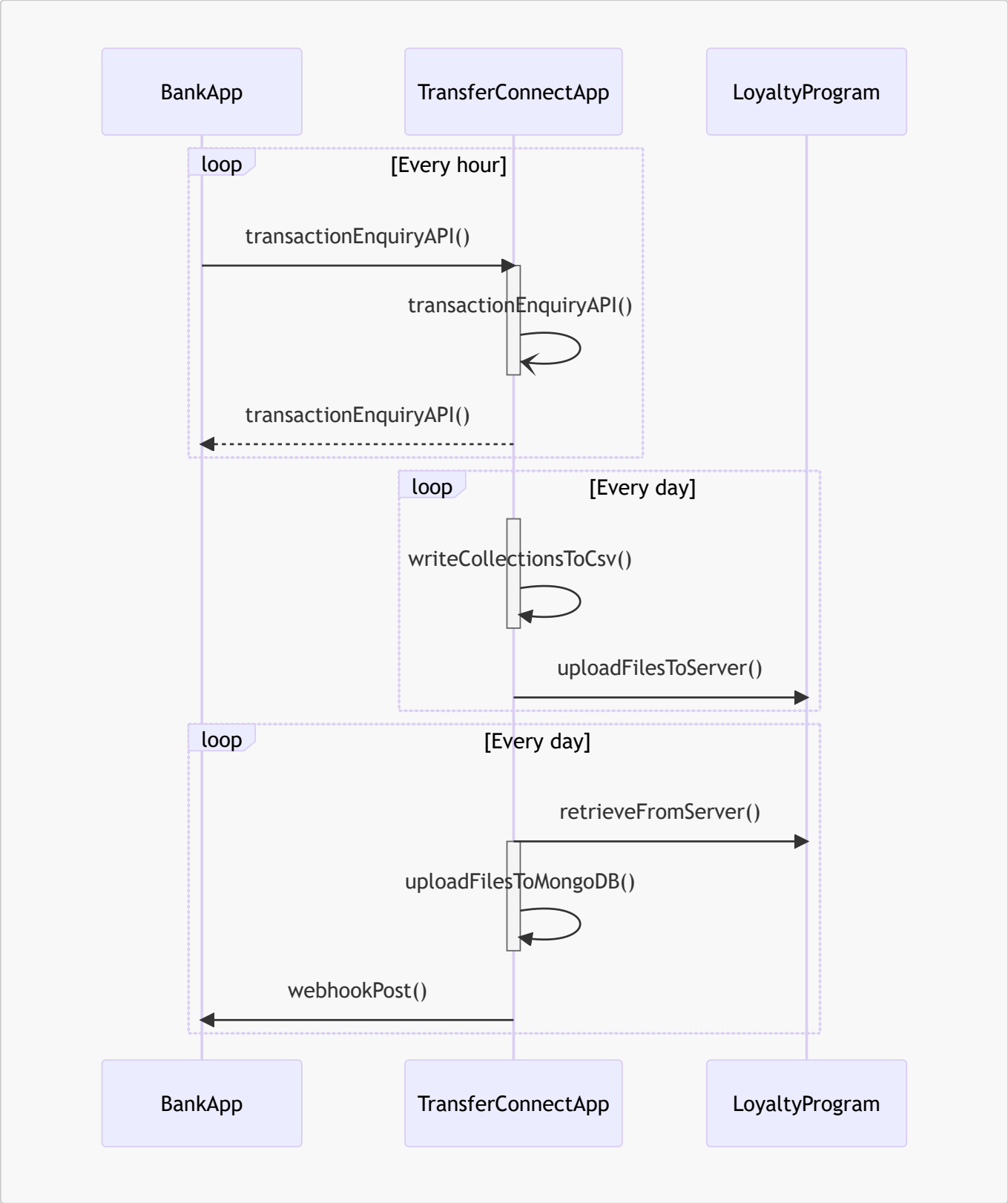
Sequence Diagrams

Transaction submission flow between Bank App client, Bank App server, Bank App database, TransferConnect server and TransferConnect database.



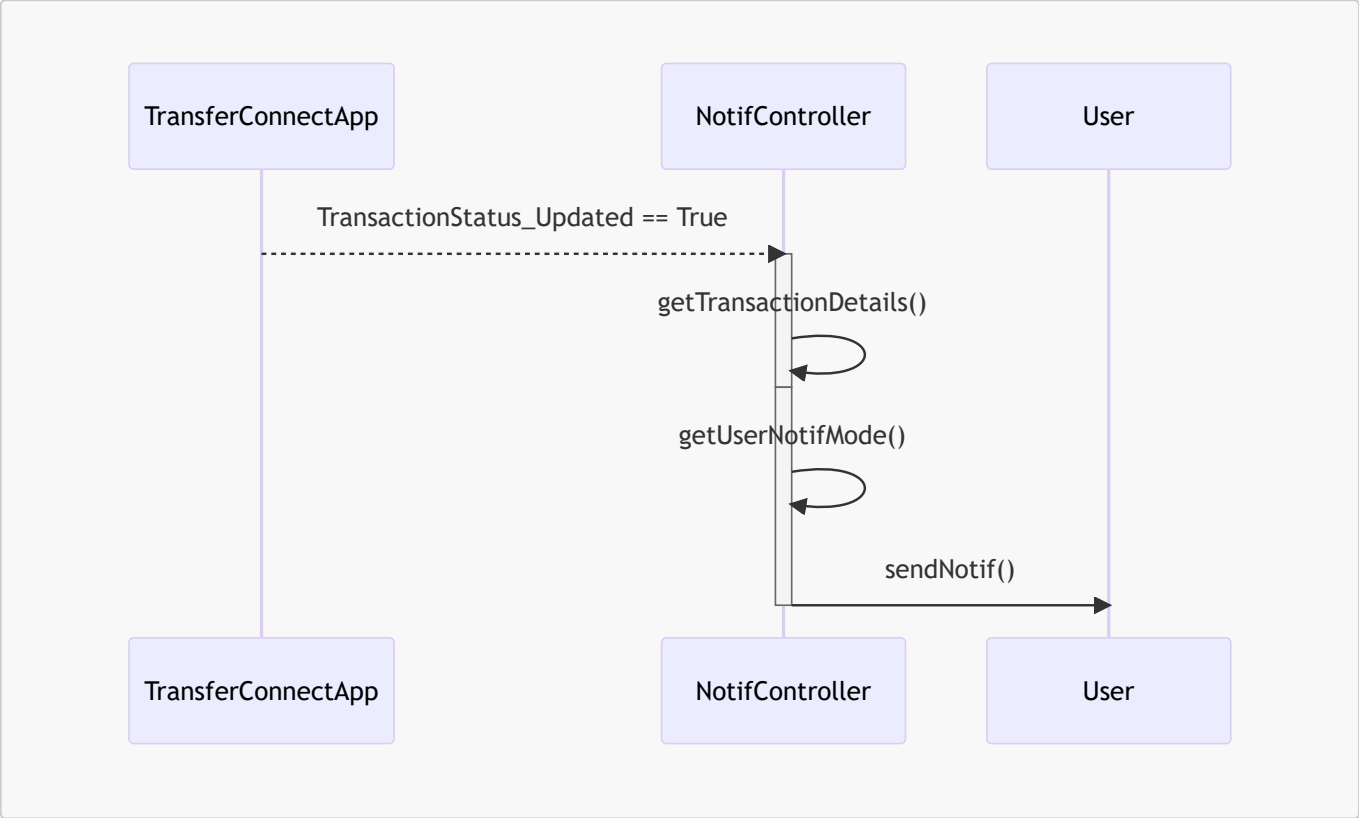
Sequence Diagrams

Transaction status update flows between Bank App, TransferConnect and Loyalty Program partners



Sequence Diagrams

Interactions between TransactionEnquiry API provided by TransferConnect App and Notification Controller in Bank App to support notifications to Bank App/Bank App User



Sequence Diagrams

Interactions between Bank App and TransferConnect App to support Bank App transaction enquiries

